

Better Target-Gradient Matching Is Not Enough: An Ordering Reversal in Targeted Instruction Selection

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Abstract

Targeted instruction selection uses query-derived signals to identify training examples useful for a target task. An improved proxy value may still fail to predict downstream utility. We introduce Directional Second-Moment Coreset (DSMC), a greedy selector with a directly measurable target-gradient discrepancy. On a controlled MMLU stress test, DSMC outperforms round-robin baselines with Llama-2-7B. The advantage reverses with Llama-3.2-3B. BBH provides the central counterexample. Across three shared query/selection draws in each of two model stacks, DSMC attains the lowest observed target discrepancy among the compared subsets and underperforms Random in every draw (−2.94 points on Llama-2-7B; −0.89 on Llama-3.2-3B). The DSMC–Random discrepancy ordering persists under post-hoc rescoring with evaluation-matched gradients and with a 256-example proxy set disjoint from every selection query. A signed first-moment selector exhibits the same qualitative reversal. The operational teacher-forced query loss can also improve with no corresponding chain-of-thought gain. In the tested final-answer-gradient targeting pipelines, lower operational first- or second-moment discrepancies therefore fail to rank downstream utility reliably.

1 Introduction

Instruction tuning (Wei et al., 2022) draws from heterogeneous pools, whereas deployment may require a narrow capability. Targeted selection uses a small query set to score candidate data through distribution alignment (Liu et al., 2024b), gradient similarity or influence (Xia et al., 2024; Wang et al., 2025), target subspaces (Min et al., 2026), or task-metric signals (Wang et al., 2025; Wu et al., 2025). We ask a narrower question about attained proxy values. Recent work studies which representations and selectors are predictive on average (Nayak et al., 2026). Our focus is whether a measured discrepancy ordering reliably ranks the utility of the compared subsets. For a fixed candidate pool, budget, training procedure A , and family of compared subsets, call the proxy order-preserving when

$$D(S_1, Q) < D(S_2, Q) \implies U(A(S_1)) \geq U(A(S_2)). \quad (1)$$

Equation 1 serves as a diagnostic definition for this paper. Useful selectors may satisfy only weaker average predictive relationships. Our experiments examine the stronger interpretation often attached to an attained proxy improvement: that it ranks the compared subsets correctly.

A downstream score alone cannot distinguish failure to attain a proxy from failure of the proxy itself. We address this ambiguity with a measurable instrument. For unit-normalized projected gradients u , represent a set by

$$M_P = \mathbb{E}_{u \sim P}[uu^\top]. \quad (2)$$

Under the quadratic kernel $k_2(u, v) = (u^\top v)^2$, empirical maximum mean discrepancy (MMD) is exactly the squared Frobenius distance between directional second moments (Gretton et al., 2012; Muandet et al., 2017). Directional Second-Moment Coreset (DSMC) greedily reduces this discrepancy using its exact one-step marginal, letting us verify target-geometry movement independently of utility.

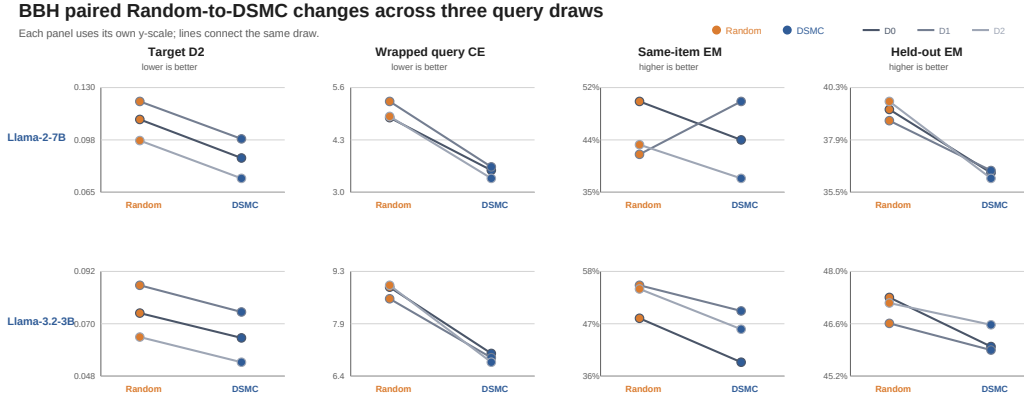


Figure 1: Target-gradient alignment and task utility diverge on BBH in both model stacks. Each line connects Random and DSMC for the same query draw; panels use separate vertical scales. Lower values are preferable for D_2 and wrapped query cross-entropy, whereas higher values are preferable for exact match. DSMC is lower in every geometry, query-loss, and held-out exact-match pair. Same-item exact match is noisier on Llama-2 but lower on average, and is lower in all three Llama-3.2 pairs. These are measured associations; causal direction remains unresolved.

The method-level results vary across stacks. On Llama-2-7B MMLU at the 5% budget, DSMC outperforms First-RR and Second-RR by 1.55 and 0.88 points, with wins in all ten target-direction cells. An internally pre-specified Llama-3.2-3B check reverses these differences (−0.18 and −0.31 points over five paired blocks). DSMC and Random remain nearly tied on both stacks. These results motivate using DSMC as a diagnostic instrument. Its performance as a general-purpose selector remains unsettled.

The BBH geometry–utility reversal has the same direction in both model stacks. Across three query/selection draws, DSMC attains the lowest operational second-moment discrepancy and loses to Random in every draw. The mean DSMC–Random gap is −2.94 percentage points on Llama-2-7B and −0.89 on Llama-3.2-3B. Every BBH SFT mean lies below its no-SFT reference. This experiment compares equally trained subsets within an observed negative-transfer regime. The discrepancy ordering also persists on a 256-example proxy set disjoint from every selection query.

We also separate targeting loss, selection discrepancy, and downstream exact match. On the 64 query items that define the signal, post-SFT teacher-forced loss decreases while a chain-of-thought generation diagnostic fails to improve. The same-item result weakens a purely query-to-test explanation. The behavior of both this diagnostic and bare-serialization loss varies by model stack, limiting the conclusion to the tested loss and serialization.

Contributions.

- We introduce DSMC, an exact-marginal greedy coreset construction with a directly measurable directional second-moment target discrepancy.
- MMLU yields opposite DSMC–round-robin rankings across the two model stacks: an advantage on Llama-2-7B and a reversal in the internally pre-specified Llama-3.2-3B check.
- On BBH, DSMC attains the lowest observed D_2 among the compared subsets in every draw yet underperforms Random in every draw on both model stacks. When the frozen subsets are rescored with evaluation-matched or query-disjoint proxy gradients, DSMC remains closer than Random in every draw. Same-item, parser, and task diagnostics constrain several explanations, while the causal mechanism remains open.

2 Target Matching with Directional Second Moments

2.1 Targeted instruction selection

Let $\mathcal{C} = \{x_i\}_{i=1}^N$ be a candidate instruction pool and $\mathcal{Q} = \{q_j\}_{j=1}^M$ a small target query set. Given a budget K , the goal is to select $\mathcal{S} \subset \mathcal{C}$, $|\mathcal{S}| = K$, such that supervised fine-tuning on $\mathcal{S}$ improves held-out target performance.

Following gradient-based targeted selection (Xia et al., 2024; Nayak et al., 2026; Mirza-soleiman et al., 2020; Killamsetty et al., 2021), we compute per-example gradients at a shared warm-up checkpoint and project them to a reusable low-dimensional datastore. Following the optimizer-aware LESS protocol, candidates and queries use role-specific transforms before entering a common projected directional space:

$$u_{\mathcal{C}}(x) = \frac{\Pi A_{\text{Adam}} g(x)}{\|\Pi A_{\text{Adam}} g(x)\|_2}, \quad u_{\mathcal{Q}}(q) = \frac{\Pi g(q)}{\|\Pi g(q)\|_2}. \quad (3)$$

Here Π is a shared Rademacher projection and A_{Adam} denotes the candidate-side optimizer-aware transform. The MMD identity below is exact for these unit vectors in the common projected space. Candidate and query features use different transforms, so the construction describes an operational push-forward geometry. Symmetric raw-gradient matching remains outside the experiment.

2.2 Why a directional second moment?

First-order mean-gradient matching represents a target by $\mathbb{E}[u]$. This signed mean can cancel when useful examples occupy opposite directions, even if they share the same underlying orientation. We instead use the non-centered directional second moment

$$M_P = \mathbb{E}_{u \sim P}[uu^\top]. \quad (4)$$

Because $uu^\top = (-u)(-u)^\top$, this representation captures orientation and discards sign. We call it a directional second moment: the vectors are normalized, and the mean is not subtracted as it would be for covariance.

Consider the positive semidefinite kernel

$$k_2(u, v) = (u^\top v)^2. \quad (5)$$

Since $\langle uu^\top, vv^\top \rangle_F = (u^\top v)^2$, the biased empirical MMD between a selected set $\mathcal{S}$ and query set $\mathcal{Q}$ is exactly (Gretton et al., 2012; Muandet et al., 2017)

$$\begin{aligned} \text{MMD}_{k_2}^2(\mathcal{S}, \mathcal{Q}) &= \left\| \frac{1}{|\mathcal{S}|} \sum_{s \in \mathcal{S}} u_s u_s^\top - \frac{1}{|\mathcal{Q}|} \sum_{q \in \mathcal{Q}} u_q u_q^\top \right\|_F^2 \\ &\equiv D_2(\mathcal{S}, \mathcal{Q}). \end{aligned} \quad (6)$$

This algebraic identity makes the empirical second moment directly measurable. The quadratic kernel is non-characteristic, so D_2 does not identify the full gradient distribution (Sriperumbudur et al., 2011; Muandet et al., 2017).

2.3 Exact greedy coreset construction

For a candidate x , define target relevance and accumulated selected-set similarity as

$$r_{\mathcal{Q}}(x) = \frac{1}{|\mathcal{Q}|} \sum_{q \in \mathcal{Q}} k_2(u_x, u_q), \quad r_{\mathcal{S}}(x) = \sum_{s \in \mathcal{S}} k_2(u_x, u_s). \quad (7)$$

At step $m = |\mathcal{S}|$, the exact one-step marginal for minimizing the biased MMD selects

$$x^* = \arg \max_{x \notin \mathcal{S}} \left[r_{\mathcal{Q}}(x) - \frac{r_{\mathcal{S}}(x) + k_2(u_x, u_x)/2}{m+1} \right]. \quad (8)$$

For unit vectors $k_2(u_x, u_x) = 1$. The first term rewards query alignment and the second penalizes selected-set self-similarity. We use this deterministic exact marginal throughout. The analysis invokes neither stochastic-greedy approximations nor submodularity guarantees. The construction is related to kernel herding and greedy kernel-mean approximation (Chen et al., 2010; Bach et al., 2012). Equation 8 is the exact marginal of our finite-sample objective.

From local alignment to downstream utility. For one small SGD step on the same query loss, $L_Q(\theta - \eta g_S) - L_Q(\theta) \approx -\eta g_Q^\top g_S$ motivates alignment locally. Our pipeline additionally applies role-specific transforms, greedy construction, multi-step Adam in a LoRA subspace, generation, and exact match. Equation 1 would require this chain to preserve subset order. The signed first-order Taylor relation does not derive D_2 as a utility surrogate or predict the full pipeline. The observed reversals remain compatible with the local approximation.

Role in this paper. DSMC provides a controlled intervention on target geometry. A lower D_2 with no utility gain isolates proxy quality from optimization failure. We use DSMC for this diagnostic role. Its broader selector performance falls outside the present study.

3 Experimental Design

3.1 Candidate pool, features, and training

The candidate pool is the processed pool released with LESS (Xia et al., 2024): 270,679 examples from Flan v2 (Longpre et al., 2023) (100,000), its CoT mixture (100,000), Dolly (Databricks, 2023) (15,011), and OASST1 (Köpf et al., 2023) (55,668). For each model stack, we construct a candidate-gradient datastore from a LoRA (Hu et al., 2022) warm-up checkpoint. Gradients are projected to 8,192 dimensions with a Rademacher projection (seed 123), unit-normalized, and cached separately for each query draw. The selected indices are materialized into static subsets. Every downstream comparison restarts from the corresponding base model. The warm-up run supplies features only. We study Llama-2-7B and Llama-3.2-3B model stacks (Touvron et al., 2023; Grattafiori et al., 2024; Meta AI, 2024), implemented through LLaMA-Factory (Zheng et al., 2024).

Downstream SFT uses LoRA rank 128, alpha 512, dropout 0.05 on q/k/v/o_proj, learning rate 2×10^{-5} , linear decay, 3% warm-up, bfloat16, effective batch size 128, and four epochs. The 5% budget is $K = 13,533$ (420 optimizer steps) and the 1% budget is $K = 2,707$ (84 steps). Full configurations, prompt templates, hashes, and data-processing details are provided in Appendix J.

3.2 Target families

MMLU controlled stress test. We construct ten globally disjoint 64-example query sets: five STEM-majority draws (51 STEM, 13 humanities) and five humanities-majority draws with the proportions reversed. Evaluation is balanced accuracy

$$\text{BalAcc} = \frac{1}{2}(\text{Acc}_{\text{STEM}} + \text{Acc}_{\text{Humanities}})$$

over held-out MMLU subjects (Hendrycks et al., 2021). We use this equal-category aggregate in place of the benchmark’s standard overall accuracy. Draw index is paired with SFT seed {42, 1, 2, 3, 4}. For comparisons shared across the two skew directions, the primary descriptive unit is the five direction-averaged draw-index blocks.

BBH external validation. Using the BIG-Bench Hard suite (Suzgun et al., 2023), we partition the official examples within each task family into a query reservoir and a disjoint evaluation set. The reservoir contains 1,302 examples and the frozen held-out set contains 5,209. The evaluation harness exposes 27 subtasks spanning 23 conceptual BBH task families, because two families contain three size-specific variants. Three independent 64-example samples from this per-subtask-stratified reservoir define the selection signals. Each covers 24–26 of the 27 subtasks. We report the size-weighted micro exact-match aggregate over the

custom held-out partition using a pinned chain-of-thought few-shot protocol. Each draw is crossed with two SFT seeds, which are averaged within the draw. The statistical unit is therefore the draw ($n = 3$), not the six seed-level runs.

3.3 Comparators and reporting policy

The controlled gradient baselines include first-order mean-gradient TopK (LESS-style (Xia et al., 2024)), first- and second-order relevance TopK, first- and second-order round-robin (RR) (Nayak et al., 2026), GIST whitening on the shared projected datastore (GIST-SharedProj (Min et al., 2026)), and a task-metric adaptation of NICE (NICE-MMLU-EM (Wang et al., 2025)). Their qualified names identify shared-protocol adaptations; the experiments are not official end-to-end reproductions. Target-independent controls include uniform Random- K , length-matched Random on MMLU, a sequence/label-length-matched Random control on BBH, and a shared no-SFT reference.

The study is descriptive. We report draw-level means, paired differences, and win counts. Samples of $n = 3$ or $n = 5$ blocks provide little resolution for significance testing. Sub-task breakdowns and individual SFT seeds serve as diagnostics and are not counted as independent replicates.

4 Method Gains Are Stack-Dependent on MMLU

4.1 Llama-2 positive control

We first evaluate whether second-moment matching provides a useful targeted-selection signal on MMLU. A single-seed exploratory 2×2 comparison on separate 80-example STEM- and humanities-majority targets factors representation (signed first order versus directional second order) and selection rule (relevance TopK versus MMD coreset). The second-order representation improves both selectors in both directions. Its additional MMD-over-TopK increment is 0.42–0.62 points, within the measured training noise. We treat this decomposition as a mechanism hypothesis. Appendix B reports the full exploratory comparison.

Across ten target draws at the 5% budget, DSMC beats LESS-style TopK and both RR baselines in every draw, and beats the examined GIST and NICE adaptations in nine of ten draws (Appendix B). Relative to Second-RR, the mean gain is 0.88 points and holds in all ten cells, supporting an observed coreset advantage at this budget. This positive result applies to the joint representation-selector construction in the Llama-2 MMLU setting.

The comparison with Random is less favorable. At 5%, DSMC trails Random- K by 0.01 points on average and wins two of five direction-averaged blocks. Length-matched Random yields the same qualitative result. At 1%, the gap widens to -0.77 points, with no query-direction cell favoring DSMC. The MMD gain over Second-RR also contracts to 0.17 points, with five of ten cells favoring DSMC.

Most targeted methods fall below no-SFT at both budgets. At 1%, DSMC (40.17) and Random (40.95) both exceed no-SFT (40.03), while DSMC is closer to the leave-one-draw-out balanced target geometry in all ten draws. The discrepancy and utility rankings therefore reverse even when both subsets improve over the base model. This deliberately skewed stress test complements the same-family BBH validation. In an equal-step sensitivity, repeating the 1% subsets to 420 steps lowers both methods to approximately 38.3% (Appendix F).

4.2 Internally pre-specified cross-stack check

The cross-stack check finds no transferable DSMC advantage. Mean balanced accuracy is 47.95 for Second-RR, 47.82 for First-RR, 47.64 for DSMC, and 47.52 for Random. The no-SFT reference is 49.47. All four methods lie within 0.44 points and below base. As Table 1 shows, DSMC’s Llama-2 advantage over both RR arms reverses on Llama-3.2, while its difference from Random stays near zero on both stacks. The BBH geometry-utility result below has greater cross-stack consistency.

Table 1: Cross-stack MMLU comparison at the 5% budget. DSMC outperforms both RR baselines in all five Llama-2 blocks. Both comparisons reverse on Llama-3.2-3B. DSMC and Random have nearly tied observed means on both stacks. Win counts use the same five direction-averaged draw-index blocks for both stacks.

Stack	DSMC – First-RR	DSMC – Second-RR	DSMC – Random
Llama-2-7B	+1.55 (5/5)	+0.88 (5/5)	−0.01 (2/5)
Llama-3.2-3B	−0.18 (2/5)	−0.31 (1/5)	+0.12 (3/5)

Table 2: Held-out BBH exact match. Means first average the two SFT seeds within each of three query/selection draws. All method means are below the corresponding no-SFT reference. DSMC loses to Random in every draw on both model stacks, while achieving the lowest target second-moment discrepancy among the compared subsets in every draw.

Stack	Method	Draw 0	Draw 1	Draw 2	Mean	Δ vs. no-SFT
Llama-2-7B	No-SFT		39.64			–
	Random- K	39.30	38.78	39.67	39.25	−0.39
	First-RR	36.24	37.61	37.61	37.15	−2.49
	Second-RR	36.80	37.00	36.80	36.87	−2.77
	DSMC	36.36	36.47	36.10	36.31	−3.33
Llama-3.2-3B	No-SFT		47.11			–
	Random- K	47.32	46.61	47.17	47.03	−0.08
	First-RR	46.11	46.52	47.04	46.56	−0.55
	Second-RR	46.62	46.23	47.18	46.68	−0.44
	DSMC	45.98	45.88	46.57	46.14	−0.97

5 External Reversal on BBH

MMLU uses deliberately skewed queries, which may be unrepresentative of its balanced evaluation. BBH draws queries and evaluation examples from the same task families, removing this specific mismatch. The geometry–utility reversal persists (Table 2). On Llama-2-7B, Random reaches 39.25% micro exact match and DSMC reaches 36.31%, a −2.94 point paired gap with zero of three draws favoring DSMC. Both RR variants also outperform DSMC. Together with Table 1, the pattern indicates that DSMC’s favorable MMLU ranking does not extend across tasks or model stacks.

The same direction appears at smaller magnitude on Llama-3.2-3B: Random is best at 47.03%, and DSMC is last among the four shared arms at 46.14%. Table 3 summarizes the draw-level contrasts and descriptive intervals. The direction repeats across stacks, but the Llama-3.2 interval includes zero. With three draws, an all-same-sign two-sided sign test has $p = 0.25$. All methods on that stack remain within one point of no-SFT.

All SFT means are below no-SFT. On Llama-2, Random is only 0.39 points below the shared base reference, while the target-aware arms are 2.49–3.63 points below it (the lower endpoint includes the LESS-style arm omitted from Table 2). On Llama-3.2, Random is 0.08 points below base and DSMC 0.97 below. These shared reference lines are not replicated treatments, so we report observed means without an inferential negative-transfer claim. Under the same Llama-2 pipeline, DSMC and Random are slightly above no-SFT on MMLU at 5%, and Random is 0.92 points above it at 1%. The same training pipeline can therefore yield positive transfer.

Matched-format control. On Llama-2, constraining Random to match DSMC’s coarse sequence-length $\times$ loss-bearing-label-length histogram lowers Random from 39.25% to 38.05%, but still leaves it 1.74 points above DSMC. This is consistent with instruction-format or correlated provenance composition contributing to the gap. Because the control changes several correlated properties simultaneously, it does not isolate a causal contribution. Full results are in Appendix F.

Table 3: Draw-level BBH contrasts. Paired DSMC–Random exact-match differences (percentage points) after averaging the two SFT seeds within each query/selection draw. Intervals are descriptive t intervals over three draws, rather than confirmatory inference. Both stacks reuse the same query draws.

Stack	Draw 0	Draw 1	Draw 2	Mean	95% interval
Llama-2-7B	−2.94	−2.31	−3.57	−2.94	[−4.50, −1.38]
Llama-3.2-3B	−1.34	−0.73	−0.60	−0.89	[−1.88, +0.10]

6 Diagnosing the Geometry–Utility Reversal

6.1 DSMC attains the lowest observed target discrepancy

For every BBH draw, we compute the operational discrepancy used by DSMC,

$$D_2(\mathcal{S}, \mathcal{Q}_d) = \|M_{\mathcal{S}} - M_{\mathcal{Q}_d}\|_F^2. \quad (9)$$

On Llama-2, mean D_2 is 0.0863 for DSMC and 0.1098 for Random; on Llama-3.2 it is 0.0644 and 0.0752. DSMC has the lowest D_2 among the compared subsets in every draw of both stacks and lower downstream utility than Random in those same draws. This paired 3/3 pattern is the primary geometry result. Appendix C reports descriptive rank correlations.

The selector achieves its intended discrepancy ordering relative to the compared subsets. This statement concerns the attained greedy solution. It does not establish global optimality. On MMLU, DSMC is also closer than Random to a leave-one-draw-out balanced reference in all ten draws at both budgets (Appendix C), despite the skewed selection queries.

Query-disjoint proxy geometry. We post-hoc score frozen subsets against 256 examples sampled after excluding all selection queries, under a protocol fixed before proxy-gradient extraction. Without reselection or SFT, DSMC remains closer than Random under D_2 , and First-RR under signed D_1 , in all three draws on both stacks. Both retain lower downstream utility. The same ordering therefore holds when geometry is measured on examples excluded from selection (Appendix D).

Sign-sensitive first-moment diagnostic. The quadratic kernel is invariant to $u \mapsto -u$, so we also examine the signed first-moment discrepancy

$$D_1(\mathcal{S}, \mathcal{Q}) = \|\mathbb{E}_{\mathcal{S}}[u] - \mathbb{E}_{\mathcal{Q}}[u]\|_2. \quad (10)$$

First-RR is closer than Random under D_1 in all three draws on both stacks, yet its held-out BBH mean is lower than Random by 2.10 points on Llama-2 and 0.48 points on Llama-3.2. The observed alignment–utility reversal is also present for this signed first-moment discrepancy. Full draw-level values appear in Appendix C. The mean gaps (0.471 vs. 0.591 and 0.459 vs. 0.498) are substantial. This diagnostic covers one sign-sensitive representation and does not characterize all such representations.

Evaluation-matched target serialization. To test whether the DSMC minimum is induced by the chat serialization used for target extraction, we re-extract only target gradients under the bare evaluation serialization and rescore frozen subsets. Despite low cross-context row cosine (0.25–0.36), DSMC retains the lowest D_2 in all draws on both stacks while losing to Random downstream. The full Llama-3.2 ranking changes in every draw. The DSMC–Random counterexample survives this serialization change, although the complete target-space ranking is serialization-dependent (Appendix E).

6.2 The defining query loss improves

Gradient-based targeting constructs query gradients from a differentiable final-answer objective. We recompute its teacher-forced cross-entropy on the same items and with the exact wrapped serialization used by each stack. On Llama-2, all four target-aware methods

reduce this loss by 0.68–1.40 nats, while Random increases it by 0.33. On Llama-3.2, the target-aware methods reduce it by 3.19–3.40 nats, compared with 1.37 for Random. The second stack shows a magnitude separation, while Llama-2 also separates the methods by sign. Across both stacks, the targeted methods improve the loss that defines the query gradients more strongly.

6.3 Same-item task utility diverges from query loss

Comparing query loss to held-out exact match changes both metric and examples. To hold the examples fixed, we run a chain-of-thought generation diagnostic on the same 64 query items. The Llama-2 arm uses the pinned query-task harness. The Llama-3.2 arm uses the same bare prompts with a custom greedy generator, a 512-token cap, and closely matched answer extraction. For Llama-2, every target-aware method’s mean query exact match falls relative to base (−3.65 to −6.51 points) despite the substantial cross-entropy reductions. On Llama-3.2, all three targeted methods again have negative mean changes, with DSMC falling by 6.51 points; Random changes by +1.30 points. One item corresponds to 1.56 points, making each 64-item draw coarse. The custom second-stack evaluator also makes those values approximate, although their mean separation agrees with the held-out ranking.

The task metric also fails to improve on the examples that define the targeting signal, which weakens a purely query-to-test generalization account. Other forms of overfitting and objective mismatch remain possible, and this diagnostic does not identify a mechanism.

Task-level breadth. The draw remains the primary experimental unit. A secondary 27-subtask breakdown shows that DSMC trails Random on 23 subtasks and leads on four for Llama-2 (macro difference −2.89 points). On Llama-3.2 it trails on 17 and leads on ten (macro difference −1.10). A descriptive bootstrap over subtasks gives intervals of [−4.68, −1.41] and [−2.57, +0.27] points, respectively. The Llama-2 gap is broad across subtasks; the smaller Llama-3.2 effect is more heterogeneous. Regrouping size variants into the 23 conceptual BBH families yields the same qualitative pattern: 20/23 and 14/23 families favor Random.

Parser-validity audit. A post-hoc audit of all frozen generations finds invalid-rate contrasts of +0.03 points on Llama-2 and −0.78 on Llama-3.2. DSMC remains below Random conditional on standard-valid outputs (−3.13, −1.36 points) and under frozen conservative recovery (−3.08, −0.84), so parsing does not account for the gap (Appendix G).

6.4 Serialization changes the loss diagnostic

The targeting loss uses each stack’s LLaMA-Factory wrapper (Zheng et al., 2024), whereas BBH generation uses a pinned bare evaluation context implemented through the language-model evaluation harness (Biderman et al., 2024). Recomputing final-answer cross-entropy on that bare context yields different behavior across stacks. On Llama-2, all methods increase bare-context CE, with smaller increases for the targeted methods. On Llama-3.2, target-aware methods decrease it and Random increases it. These results tie the loss-based finding to the tested serialization and model stack. They show that the targeting loss can decrease on its defining examples while chain-of-thought exact match fails to improve. The evidence does not support a general claim about cross-entropy and generation.

7 Related Work

Instruction data selection. General-purpose instruction selection commonly scores data for quality, difficulty, and diversity. DEITA jointly studies complexity, quality, and diversity for data-efficient alignment (Liu et al., 2024a). Get More for Less studies selection for the warm-up stage before target fine-tuning (Kang et al., 2024). TSDS uses a small target set to define an optimal-transport distribution-alignment objective (Liu et al., 2024b). Our focus is on testing whether attaining a better query-derived proxy value is sufficient for utility.

Distribution matching and gradient coresets. MMD and kernel mean embeddings define set-level distribution discrepancies (Gretton et al., 2012; Muandet et al., 2017). Characteristic-kernel results specify when these discrepancies identify distributions (Sriperumbudur et al., 2011). Kernel herding greedily approximates a target kernel mean (Chen et al., 2010; Bach et al., 2012), while CRAIG and GRAD-MATCH use gradient geometry for data-efficient training (Mirzasoleiman et al., 2020; Killamsetty et al., 2021). We use the established quadratic-kernel identity and greedy moment matching as a measurable intervention for testing the downstream value of stronger target-gradient alignment.

Targeted and influence-based selection. LESS builds reusable low-dimensional gradient datastores and uses optimizer-aware gradient similarity (Xia et al., 2024). NICE links influence estimation to non-differentiable task metrics through policy gradients (Wang et al., 2025). GIST scores candidates in a target-gradient subspace (Min et al., 2026), and GIO optimizes a scalable gradient-information objective (Everaert & Potts, 2024). A controlled study that separates representations from selectors finds useful gradient signals but no uniformly dominant selector (Nayak et al., 2026). Our study conditions on the attained geometry and asks whether it ranks subset utility.

Surrogate mismatch and Random. ROSE and NICE motivate task-metric-aware selection from loss-reward mismatch (Wu et al., 2025; Wang et al., 2025), while controlled studies find that complex selectors may not consistently beat Random (Diddee & Ippolito, 2025; Xia et al., 2025). Our contribution verifies proxy attainment and traces its failure through query-disjoint geometry and same-item metrics.

8 Limitations

The study covers two Llama-family stacks, one candidate pool, and two target families. Each stack changes model, tokenizer, and serialization together. BBH has three draws reused across stacks, and the 3B effect is small. On Llama-3.2 MMLU, all methods are within 0.44 points and below no-SFT. MMLU also couples draw index to seed, so five direction-averaged blocks carry shared comparisons.

Candidates use Adam-aware features and queries use raw SGD gradients. Our claims concern this operational push-forward geometry. Symmetric raw-gradient matching remains untested. Evaluation-matched rescoring leaves the transform asymmetry intact, and the finer Llama-3.2 ranking changes in every draw.

Baselines are controlled adaptations, and contamination checks are lexical. The diagnostics leave the causal mechanism open: the proxy rescores frozen subsets, and the parser audit tests extraction. BBH targets a wrapped final-answer loss and evaluates bare-context CoT generation, which limits the claim to this pipeline. Protocol files document internally pre-specified decisions. The anonymous snapshot cannot independently establish chronology.

9 Conclusion

DSMC’s method-level advantage varies across stacks, whereas the geometry-utility reversal has the same direction in both. The positive-transfer MMLU 1% comparison and the negative-transfer BBH regime each contain subsets whose lower operational first- or second-moment gradient discrepancy coincides with lower downstream utility. Improvements in a query-derived objective therefore require separate validation against the downstream metric.

AI use statement

Generative AI tools were used to summarize and cross-check existing literature, inspect repository artifacts, suggest paper structure and wording, draft and edit parts of the manuscript, and assist with software implementation and analysis scripts. Earlier in the research workflow, generative AI tools also provided feedback on experimental design and result interpretation. The authors reviewed all AI-assisted text, code, and claims against

the underlying implementations, configurations, manifests, and result files. All reported numerical results were computed from experiment artifacts. No reported result was generated by an AI system. The authors take responsibility for the final content, including all AI-assisted material.

Ethics statement

This work studies data selection for publicly available language-model benchmarks and does not involve human-subject data collection. The main risks are those shared by instruction tuning, including inherited dataset biases, benchmark contamination, and overgeneralization from a limited experimental scope. We report negative results, adapted-baseline scope, and contamination audit limits explicitly to reduce the risk of overstating practical conclusions.

Reproducibility statement

Appendix J documents the data composition, target-draw construction, feature protocol, selection rules, training hyperparameters, evaluation pins, statistical units, and integrity checks. The supplementary material will include anonymized code, exact run configurations, frozen manifests and hashes, paper tables generated from committed result files, and the internally pre-specified protocol documents for the BBH and second-model experiments.

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A Method Derivations

A.1 MMD identity

Let $\widehat{M}_S = |\mathcal{S}|^{-1} \sum_{s \in \mathcal{S}} u_s u_s^\top$ and define $\widehat{M}_Q$ analogously. Expanding the squared Frobenius norm gives

$$\begin{aligned} \|\widehat{M}_S - \widehat{M}_Q\|_F^2 &= \frac{1}{|\mathcal{S}|^2} \sum_{s, s'} \langle u_s u_s^\top, u_{s'} u_{s'}^\top \rangle_F \\ &\quad + \frac{1}{|\mathcal{Q}|^2} \sum_{q, q'} \langle u_q u_q^\top, u_{q'} u_{q'}^\top \rangle_F \\ &\quad - \frac{2}{|\mathcal{S}||\mathcal{Q}|} \sum_{s, q} \langle u_s u_s^\top, u_q u_q^\top \rangle_F \end{aligned} \quad (11)$$

$$= \frac{1}{|\mathcal{S}|^2} \sum_{s, s'} (u_s^\top u_{s'})^2 + \frac{1}{|\mathcal{Q}|^2} \sum_{q, q'} (u_q^\top u_{q'})^2 - \frac{2}{|\mathcal{S}||\mathcal{Q}|} \sum_{s, q} (u_s^\top u_q)^2. \quad (12)$$

This is the biased empirical MMD under k_2 .

A.2 Greedy marginal

Terms depending only on $\mathcal{Q}$ can be dropped. Comparing the objective after adding x to a set of size m yields Equation 8 after positive rescaling. The implementation maintains $r_S(x)$ incrementally and masks selected or invalid rows. Zero and non-finite feature rows are rejected before selection.

Table 4: Exploratory Llama-2 MMLU representation $\times$ selector comparison. Balanced accuracy uses one seed on separate legacy 80-example STEM- and humanities-majority targets, not the ten 64-example draws used for the main comparisons. The DSMC-Second-TopK increment is within the measured training-noise scale.

Target	1st-TopK	1st-MMD	2nd-TopK	DSMC
STEM-majority	39.77	38.47	40.68	41.10
Humanities-majority	39.31	38.24	39.92	40.54

Table 5: Llama-2 MMLU target-draw results. Mean equal-domain balanced accuracy over ten query draws (five paired draw-index blocks). The no-SFT reference is shared, not a replicate.

Method	5%	1%
No-SFT	40.03	40.03
LESS-style TopK	39.41	39.37
First-RR	38.80	38.20
Second-RR	39.48	40.00
GIST-SharedProj	38.85	39.45
NICE-MMLU-EM	39.25	39.42
DSMC	40.36	40.17
Random- K	40.37	40.95
Length-matched Random	40.42	40.80

B Additional MMLU Results

Internally pre-specified Llama-3.2 outcome. The 5% cross-stack arm completed 35/35 adapters and 36/36 evaluations with no failures. The statistical unit is five direction-averaged draw-index blocks. The protocol document records a freeze before the second-model computations. The single-snapshot anonymous artifact does not independently verify this chronology. The no-SFT balanced reference is 49.47. Mean balanced accuracy is 47.95 for Second-RR, 47.82 for First-RR, 47.64 for DSMC, and 47.52 for Random. DSMC minus First-RR is -0.18 points (2/5 blocks), minus Second-RR is -0.31 (1/5), and minus Random is $+0.12$ (3/5). The outcome matches the internally pre-specified case in which the Llama-2 method advantage fails to transfer.

Lower budget and equal-step sensitivity. On Llama-2, DSMC minus Random changes from -0.01 points at 5% to -0.77 at 1%, with 0/10 1% cells favoring DSMC. Repeating the 1% DSMC and Random subsets to the 420 steps used by the 5% arm reduces both to approximately 38.3%, about 1.8 points below the shared no-SFT reference. Additional optimizer steps leave the low-budget ordering unchanged.

C Additional Geometry Diagnostics

On MMLU, a leave-one-draw-out balanced reference gives mean D_2 values of 0.1538 for DSMC and 0.1762 for Random at 5%, and 0.1522 versus 0.1764 at 1%. DSMC is closer in all ten draws at each budget. On BBH, the descriptive Spearman correlation between D_2 and accuracy is positive in all three Llama-2 draws (0.77, 0.83, 0.89 over six methods) and all three Llama-3.2 draws (0.80, 1.00, 0.20 over four methods). These ranking samples are too small for inferential use and are reported descriptively.

When the size-specific logical-deduction and shuffled-object subtasks are regrouped into the 23 conceptual BBH families, DSMC wins/ties/loses 3/0/20 families on Llama-2 and 9/0/14 on Llama-3.2. Family-macro differences are -2.94 and -1.23 points, with descriptive task-family bootstrap intervals $[-4.94, -1.25]$ and $[-2.91, +0.26]$.

Table 6: Signed first-moment discrepancy on BBH. $D_1 = \|\mathbb{E}_S[u] - \mathbb{E}_Q[u]\|_2$ is sign-sensitive. First-RR is closer than Random in every draw, yet its downstream mean is lower on both stacks.

Stack	Draw	DSMC	First-RR	Random
Llama-2	0	0.4886	0.4763	0.5957
	1	0.4972	0.4861	0.6032
	2	0.4610	0.4516	0.5741
Llama-3.2	0	0.4404	0.4611	0.5012
	1	0.4599	0.4789	0.5179
	2	0.4081	0.4374	0.4736

Table 7: Task-level DSMC–Random heterogeneity. Win/tie/loss counts and macro statistics average SFT seeds and query draws within each of the 27 harness subtasks. Bootstrap intervals resample subtasks and describe heterogeneity across tasks. The pre-specified experimental unit remains the draw.

Stack	W/T/L	Macro Δ	Median Δ	Task-bootstrap 95%
Llama-2	4/0/23	−2.89	−1.67	[−4.68, −1.41]
Llama-3.2	10/0/17	−1.10	−0.92	[−2.57, +0.27]

D Query-Disjoint Proxy Geometry

For this post-hoc target-only analysis, the protocol was fixed before proxy gradient extraction. We remove the 185 distinct selection-query examples from the 1,302-example BBH query reservoir, then sample 256 examples by deterministic proportional stratification over the 27 harness subtasks (seed 20260827). The resulting proxy set is disjoint from every selection query. We rescore the frozen subsets without rerunning selection, training, or downstream evaluation.

Table 8: Geometry against the query-disjoint BBH proxy set. Lower is better. The frozen DSMC subset is closer than Random under D_2 in every stack–draw cell. First-RR is closer than Random under signed D_1 in every cell.

Stack	Draw	Metric	DSMC	First-RR	Second-RR	Random
Llama-2	0	D_1	0.46684	0.45508	0.50589	0.57679
		D_2	0.07067	0.07250	0.07482	0.09445
	1	D_1	0.46540	0.45579	0.50108	0.57522
		D_2	0.07069	0.07256	0.07409	0.09360
	2	D_1	0.46524	0.45562	0.50093	0.57786
		D_2	0.07067	0.07237	0.07392	0.09412
Llama-3.2	0	D_1	0.41998	0.44544	0.43849	0.48214
		D_2	0.05035	0.06087	0.05868	0.06056
	1	D_1	0.42084	0.44568	0.43871	0.48190
		D_2	0.05037	0.06083	0.05886	0.06132
	2	D_1	0.41935	0.44450	0.44258	0.48403
		D_2	0.05034	0.06230	0.06010	0.06103

The proxy comes from the held-out query reservoir and is distinct from the 5,209-example downstream evaluation partition. These results establish the geometry ordering of the frozen subsets. Selection and training against the proxy target remain untested.

E Evaluation-Matched Target-Geometry Sensitivity

This post-hoc check changes target serialization while retaining the frozen selections and adapters. Mean row cosine between operational and evaluation-matched target gradients is 0.335/0.342/0.363 for Llama-2 and 0.249/0.270/0.272 for Llama-3.2.

Table 9: Evaluation-matched D_2 for frozen BBH subsets. Lower is better. The frozen DSMC subset is closest in all six stack-draw cells, but the non-DSMC ranking changes in all three Llama-3.2 draws.

Stack	Draw	DSMC	First-RR	Second-RR	Random
Llama-2	0	0.06675	0.06830	0.07102	0.08875
	1	0.08230	0.08393	0.08586	0.10358
	2	0.05655	0.05800	0.05989	0.07831
Llama-3.2	0	0.07985	0.09134	0.08921	0.08847
	1	0.07660	0.08809	0.08619	0.08598
	2	0.07124	0.08414	0.08203	0.08051

F Sensitivity and Falsification Checks

Matched Random controls. MMLU length-matched Random gives the same qualitative target-awareness result as plain Random: DSMC shows no advantage at either budget. On Llama-2 BBH, sequence $\times$ label-length-matched Random scores 38.05, between plain Random (39.25) and DSMC (36.31). The control also changes correlated provenance and content, precluding a causal percentage decomposition of the format contribution.

Same-item and bare-context diagnostics. On Llama-2, targeted methods reduce the wrapped targeting loss by 0.68–1.40 nats but their same-item CoT exact match changes by -3.65 to -6.51 points. Under the bare evaluation context, no method improves CE. Targeted methods degrade it less than Random. On Llama-3.2, targeted methods reduce wrapped targeting loss by 3.19–3.40 nats and also reduce bare CE by 0.41–0.57, while their mean same-item CoT changes remain negative. The loss-based diagnostic is therefore specific to the tested targeting pipeline and does not characterize cross-entropy in general.

Task exposure. Per-task query frequency has negative descriptive correlations with targeted-vs-Random subtask deltas (-0.210 , -0.280 , -0.224 across the three draws). Greater query exposure provides no apparent protection for a BBH task. These descriptive correlations do not identify a causal mechanism.

G Parser-Validity Audit

We audit every saved BBH generation without regenerating outputs. Conditional EM is computed only over examples accepted by the standard harness parser. For standard-invalid outputs, a frozen conservative rule inspects only the final 500 characters and extracts a type-constrained final candidate: the last parenthesized option, boolean/yes-no token, or standalone number when applicable, otherwise the last explicit answer marker or nonempty line. Unrecovered outputs remain incorrect. This secondary rule is reported as a sensitivity analysis alongside the primary metric.

The DSMC–Random invalid-rate difference is $+0.03$ percentage points on Llama-2 and -0.78 on Llama-3.2. The corresponding conditional-EM differences are -3.13 and -1.36 points, and conservative-EM differences are -3.08 and -0.84 . The utility ordering therefore persists after accounting for parser validity. The audit does not assess semantic correctness beyond these extraction rules.

Table 10: BBH parser-validity audit. Rates first average the two SFT seeds within each draw and then the three draws. Conservative EM applies the frozen recovery rule only to standard-invalid outputs.

Stack	Method	Standard valid	Conditional EM	Conservative EM
Llama-2	DSMC	93.46	38.85	36.42
	Random	93.50	41.98	39.50
Llama-3.2	DSMC	94.42	48.87	46.77
	Random	93.64	50.23	47.61

H Baseline Implementations

LESS-style TopK uses one shared checkpoint and mean signed gradient similarity; it differs from the official multi-checkpoint LESS pipeline. GIST-SharedProj applies the paper’s target-subspace whitening construction to the shared normalized projected caches. NICE-MMLU-EM uses eight Monte Carlo samples and an exact-match task reward in the common feature protocol. These adaptations isolate representation and selection effects under controlled features and are not exact end-to-end reproductions of the official systems.

I Contamination and Prompt Audits

Against MMLU and held-out BBH, normalized exact matching finds zero candidate duplicates. Full-pool fuzzy screening finds zero pairs above Jaccard 0.3. Seven MMLU and five BBH 13-gram hits were manually reviewed and judged boilerplate false positives. The BBH few-shot audit finds no exact duplicates and discloses the observed fuzzy near-duplicates. Prompt parity, few-shot leakage, and token-truncation audits are included in the supplementary artifact. A semantic embedding nearest-neighbor audit remains outside the study, so the reported lexical screens do not constitute a complete decontamination certificate.

J Reproducibility Details

The candidate pool contains 270,679 examples. Projected gradients use dimension 8,192 and seed 123. MMLU target draws are globally disjoint. BBH uses three independent 64-example samples from a per-subtask-stratified reservoir and a 5,209-example held-out set. Frozen manifests record target files, gradient tensors, selections, materialized subsets, adapters, and evaluation-result hashes. All BBH evaluations pass 27-subtask/5,209-example integrity checks, and all Llama-3.2 MMLU evaluations cover 57 subjects. Selection-level stability was checked after target-gradient non-bit-reproducibility was observed. The worst selector changed 5 of 2,707 indices. A paper-level artifact manifest maps each major figure and table to its committed result files, analysis code, and frozen protocol.

J.1 Compute and selection overhead

The model-in-the-loop feature pipeline dominates the practical asymmetry: Random requires no warm-up or gradient datastore. The anonymous supplement will include sanitized configurations and manifests after removing cluster paths, Git metadata, user names, and identifying repository links.

Table 11: Representative measured or launch-accounted costs. Candidate datastores are reusable across target draws and tasks for a fixed model stack. SFT entries are sums of per-adapter elapsed time for jobs using eight NVIDIA H20 GPUs. Earlier pipeline entries are coarse accelerator-hour accounting and should be read as stage-level estimates.

Stage	Reuse scope	Representative cost
Warm-up checkpoint	per model stack	~1.5 GPU-h
Candidate gradient datastore	across targets	~6–8 GPU-h per stack
Three BBH target caches	per query draws	~1 GPU-h
DSMC + First/Second-RR	per three draws	~1.5 GPU-h
Random- K sampling	per subset	negligible
Llama-3.2 BBH SFT, 24 adapters	per selected subsets	3.39 h; 27.1 GPU-h
Llama-3.2 MMLU SFT, 35 adapters	per selected subsets	16.88 h; 135.0 GPU-h
Llama-2 BBH SFT, 36 adapters	per selected subsets	13.63 h; 109.1 GPU-h